

Binary

- Thresholding
- Threshold detection
- Variations
- Mathematical Morphology

Based on *A Practical Introduction to Computer Vision with OpenCV* by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

1

Thresholding

- Binary thresholding
 - for all pixels

$$g(i,j) = 1 \text{ for } f(i,j) \geq T$$

$$= 0 \text{ for } f(i,j) < T$$
 - Simple scenes?
 - Look Up Table
 - for all grey levels

$$LUT(k) = 1 \text{ for } k \geq T$$

$$= 0 \text{ for } k < T$$
 - for all pixels

$$g(i,j) = LUT(f(i,j))$$
 - Objects of interest vs. background

```
threshold(gray_image, binary_image, threshold,
          255, THRESH_BINARY);
```

Based on *A Practical Introduction to Computer Vision with OpenCV* by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

2

Thresholding

- Distinct foreground & background needed

Threshold = 10

How do we determine the best threshold?

Based on *A Practical Introduction to Computer Vision with OpenCV* by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

3

Threshold Detection

- Manual Setting
 - Issue of changing lighting
- Need to determine automatically

- For the techniques which follow:
 - Image – $f(i,j)$
 - Histogram – $h(g)$ of values in $f(i,j)$
 - Normalised Probability Distribution – $p(g) = h(g) / \sum_g h(g)$

Based on *A Practical Introduction to Computer Vision with OpenCV* by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

4

Threshold Detection – Otsu Thresholding

- Minimize the spread of the pixels...
 - Smallest within class variance

$$\sigma_w^2(T) = w_f(T)\sigma_f^2(T) + w_b(T)\sigma_b^2(T)$$

$$w_f(T) = \sum_{g=T}^{255} p(g) \quad \sigma_f^2(T) = \frac{\sum_{g=T}^{255} p(g) \cdot (g - \mu_f(T))^2}{w_f(T)}$$

$$w_b(T) = \sum_{g=0}^{T-1} p(g) \quad \sigma_b^2(T) = \frac{\sum_{g=0}^{T-1} p(g) \cdot (g - \mu_b(T))^2}{w_b(T)}$$

$$\mu_f(T) = \frac{\sum_{g=T}^{255} p(g) \cdot g}{w_f(T)} \quad \mu_b(T) = \frac{\sum_{g=0}^{T-1} p(g) \cdot g}{w_b(T)}$$
 - Largest between class variance

$$\sigma_b^2(T) = w_f(T)w_b(T)(\mu_f(T) - \mu_b(T))^2$$

Based on *A Practical Introduction to Computer Vision with OpenCV* by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

5

Threshold Detection – Otsu Thresholding

```
threshold( gray_image, binary_image, threshold,
          255, THRESH_BINARY | THRESH_OTSU );
```

Based on *A Practical Introduction to Computer Vision with OpenCV* by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

6

Variations – Adaptive Thresholding

- The adaptive thresholding algorithm is
 - Divide the image into sub-images,
 - Compute thresholds for all sub-images,
 - Interpolate thresholds for every point using bilinear interpolation.



Binary

Based on A Practical Introduction to Computer Vision with
OpenCV by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

Slide 7

7

Variations – Adaptive Thresholding

- OpenCV version:

$$\text{if } ((f(i,j) - (\sum_{a=-m,m} \sum_{b=-m,m} f(i+a,j+b)) / (2m+1)^2)) > \text{offset})$$

$$g(i,j) = 255$$

$$\text{else } g(i,j) = 0$$



```
adaptiveThreshold( gray_image, binary_image, output_value,
    ADAPTIVE_THRESH_MEAN_C, THRESH_BINARY,
    block_size, offset );
```

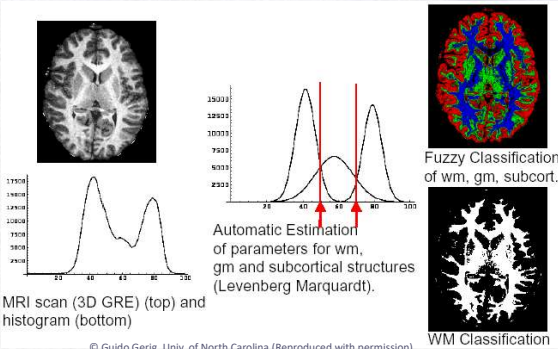
Binary

Based on A Practical Introduction to Computer Vision with
OpenCV by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

Slide 8

8

Variations – Multi-Spectral Thresholding



Binary

Based on A Practical Introduction to Computer Vision with
OpenCV by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

Slide 9

9

Cleaning Binary Images

- Cannot use normal smoothing operations
- Need to remove noisy points and smooth binary region boundaries.
- Mostly we use operations originally defined as part of “mathematical morphology” (which treats images as sets).
- Most common operations:
 - Erosion
 - Dilation
 - Opening
 - Closing

Binary

Based on A Practical Introduction to Computer Vision with
OpenCV by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

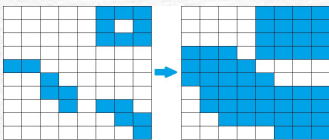
Slide 10

10

Mathematical Morphology – Dilation

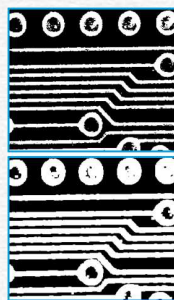
- Minkowski set addition:
 - $X \oplus B = \{ p \in \mathbb{E}^2; p = x + b, x \in X \text{ and } b \in B \}$
 - B is the structuring element
 - B is typically isotropic

- Adds pixels around borders



- Effects:

- Makes ‘regions’ bigger
- Fills small holes
- Joins close ‘regions’



Binary

Based on A Practical Introduction to Computer Vision with
OpenCV by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

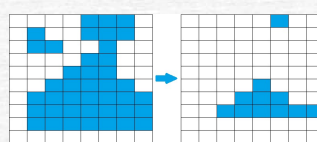
Slide 11

11

Mathematical Morphology – Erosion

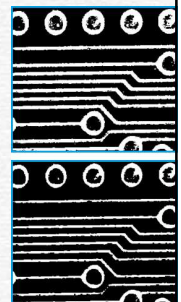
- Minkowski set subtraction:
 - $X \ominus B = \{ p \in \mathbb{E}^2; p + b \in X \text{ for every } b \in B \}$
 - Again B is typically isotropic

- Removes border pixels



- Effects:

- Makes ‘regions’ smaller
- Removes noise
- Removes narrow bridges



Binary

Based on A Practical Introduction to Computer Vision with
OpenCV by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

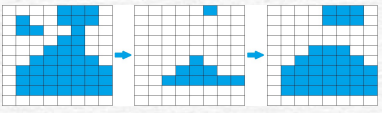
Slide 12

12

Mathematical Morphology –

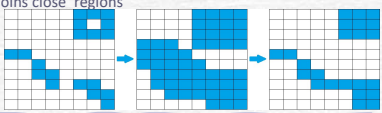
Opening: $X \circ D = (X \ominus D) \oplus D$

- Removes noise
- Removes narrow bridges
- Roughly maintains 'region' size
- Smooths shape



Closing: $X \bullet D = (X \oplus D) \ominus D$

- Fills small holes
- Joins close 'regions'
- Roughly maintains 'region' size



Based on *A Practical Introduction to Computer Vision with OpenCV* by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

Binary Slide 13

13

Mathematical Morphology – Application

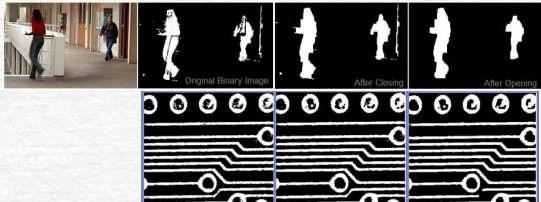
Opening: $X \circ D = (X \ominus D) \oplus D$

- Eliminates small image details

Closing: $X \bullet D = (X \oplus D) \ominus D$

Properties

- $X \circ D = (X \circ D) \circ D$ and $X \bullet D = (X \bullet D) \bullet D$



Based on *A Practical Introduction to Computer Vision with OpenCV* by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

Binary Slide 14

14

Mathematical Morphology – OpenCV Code

```
dilate( binary_image, dilated_image, Mat());
Mat structuring_element( 5, 5, CV_8U, Scalar(1) );
dilate( binary_image, dilated_image, structuring_element);

erode( binary_image, eroded_image, Mat());
Mat structuring_element( 5, 5, CV_8U, Scalar(1) );
erode( binary_image, eroded_image, structuring_element);

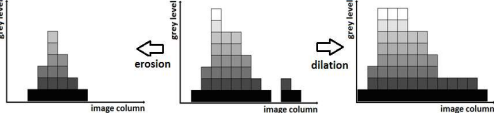
Mat five_by_five_element( 5, 5, CV_8U, Scalar(1) );
morphologyEx( binary_image, opened_image,
              MORPH_OPEN, five_by_five_element );
morphologyEx( binary_image, closed_image,
              MORPH_CLOSE, five_by_five_element );
```

Based on *A Practical Introduction to Computer Vision with OpenCV* by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

Binary Slide 15

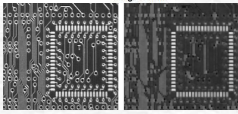
15

Mathematical Morphology – Greyscale / Colour



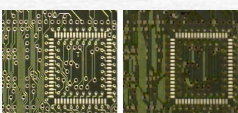
Grey scale image

- One set per grey level (g)
- All points $\geq g$
- Operations on each set separately



Colour

- One set per level per channel
- e.g. RGB image has 255×3 sets.



Based on *A Practical Introduction to Computer Vision with OpenCV* by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

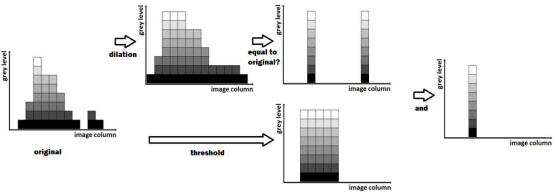
Binary Slide 16

16

Mathematical Morphology – Local maxima

Can be used to locate local maxima and minima

- Dilate & compare to the original image (to find local maxima)
- Threshold the original image
- Logical AND the results together to find 'high' local maxima



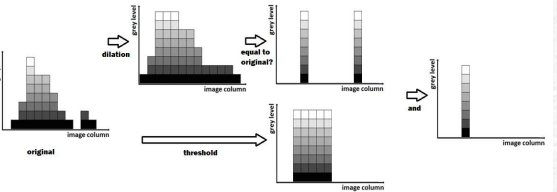
Based on *A Practical Introduction to Computer Vision with OpenCV* by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

Binary Slide 17

17

Mathematical Morphology – Local maxima

```
Mat dilated, thresholded_input, local_maxima, thresholded_8bit;
dilate( input, dilated, Mat());
compare( input, dilated, local_maxima, CMP_EQ );
threshold( input, thresholded_input, threshold, 255, THRESH_BINARY );
thresholded_input.convertTo( thresholded_8bit, CV_8U );
bitwise_and( local_maxima, thresholded_8bit, local_maxima );
```



Based on *A Practical Introduction to Computer Vision with OpenCV* by Kenneth Dawson-Howe © Wiley & Sons Inc. 2014

Binary Slide 18

18